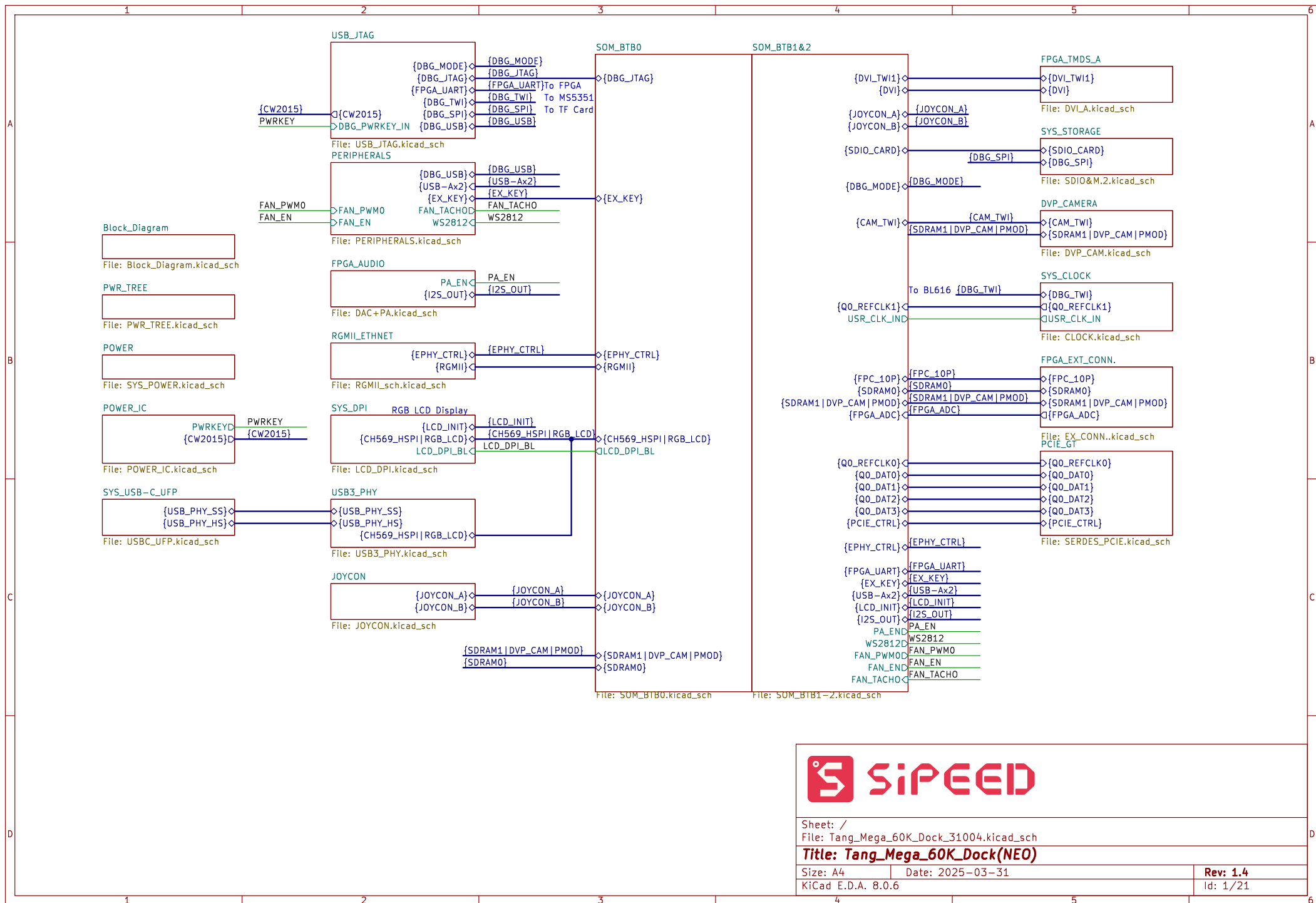
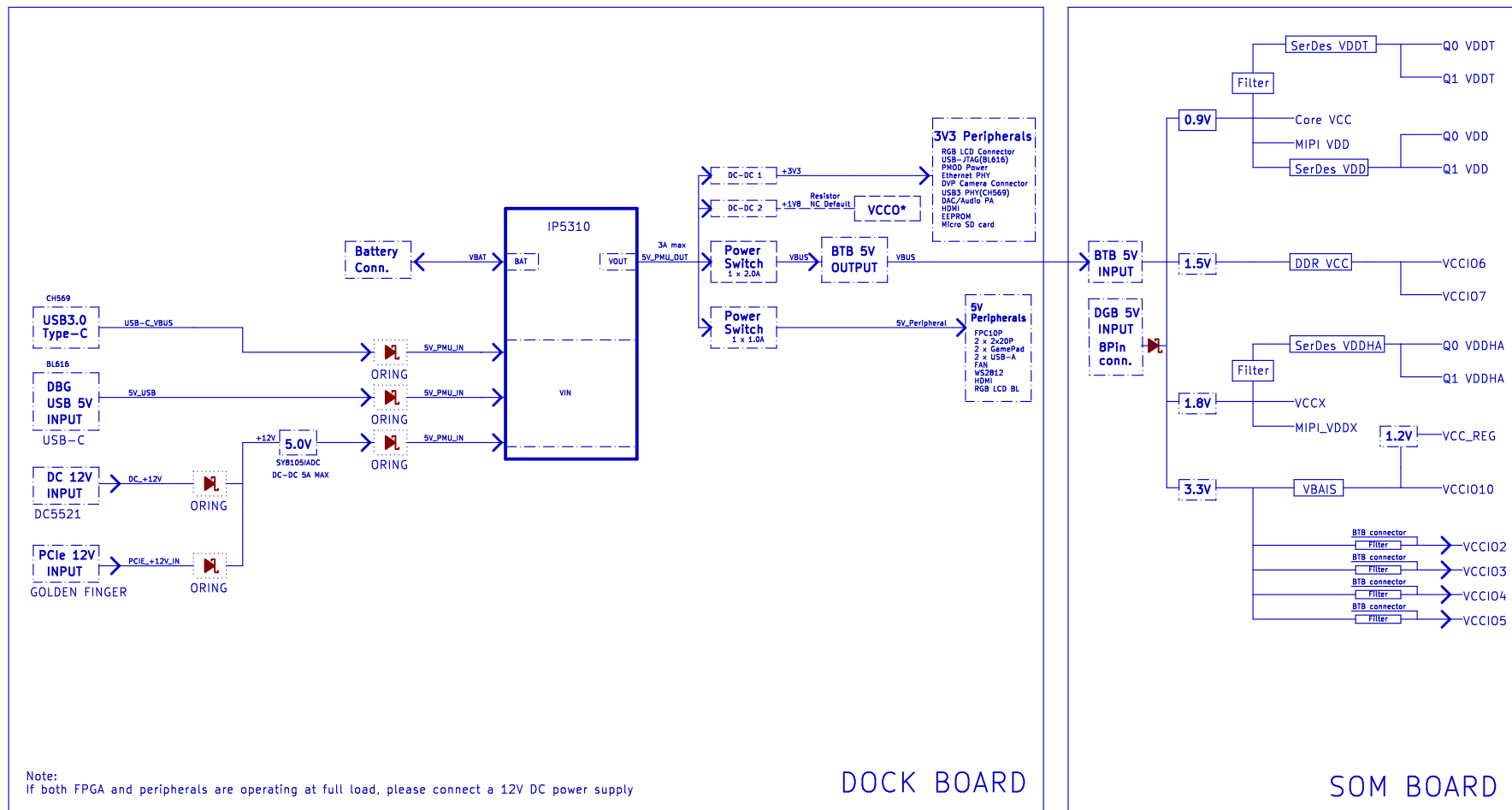






Sheet: /PWR_TREE/

File: PWR_TREE.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

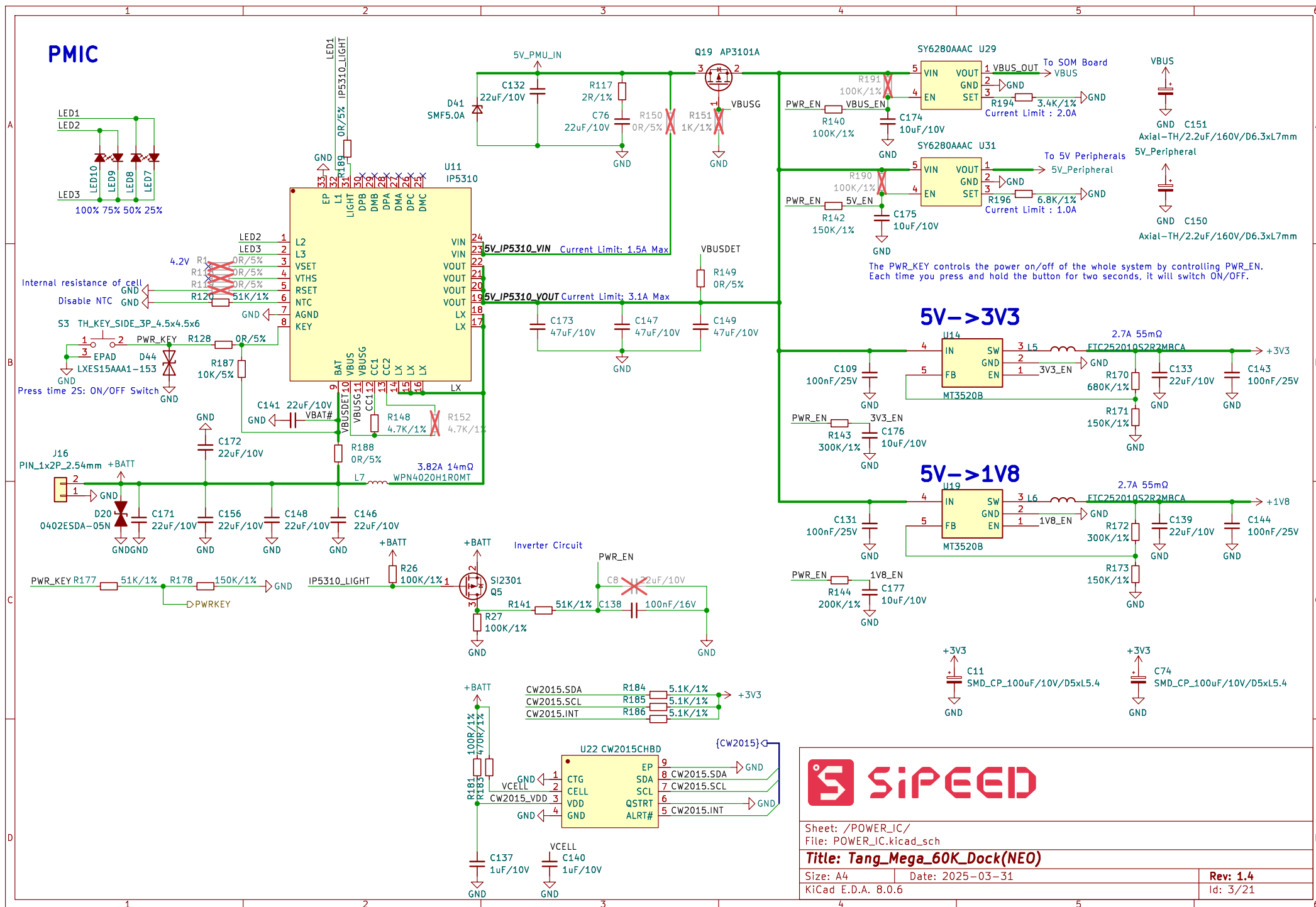
Size: A4

Date: 2025-03-31

KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 2/21



Sheet: /POWER_IC/
File: POWER_IC.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4

Date: 2025-03-31

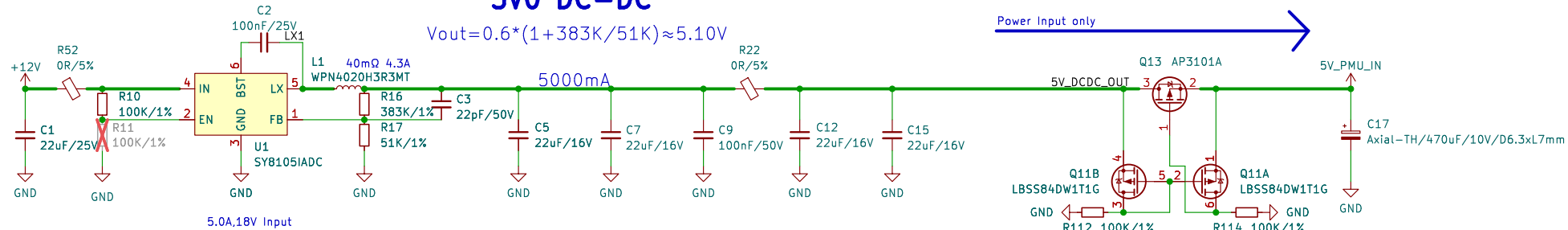
KiCad E.D.A. 8.0.6

Rev: 1.4

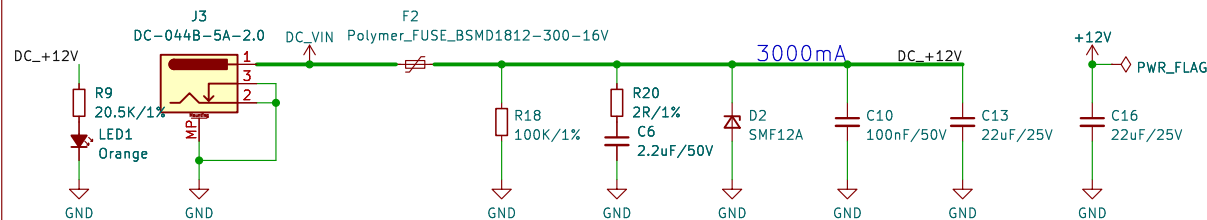
Id: 3/21

5V0 DC-DC

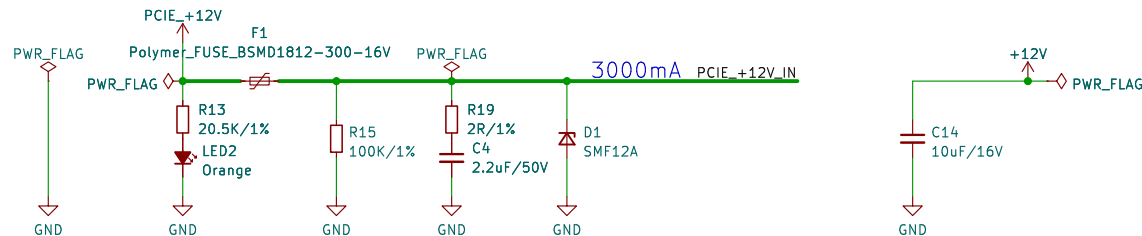
$$V_{out} = 0.6 * (1 + 383K/51K) \approx 5.10V$$



12V DC-IN



12V PCIE-IN



POWER-IN ORING



Sheet: /POWER/
File: SYS_POWER.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

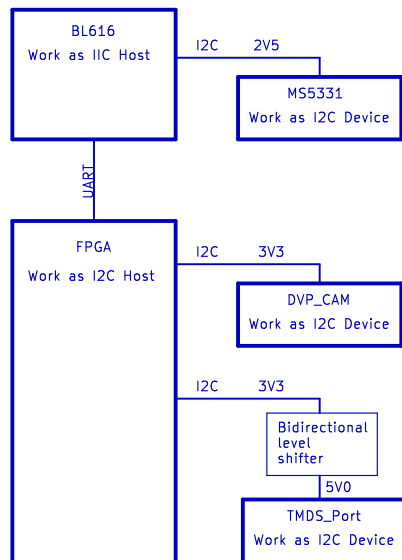
Size: A4 Date: 2025-03-31

KiCad E.D.A. 8.0.6

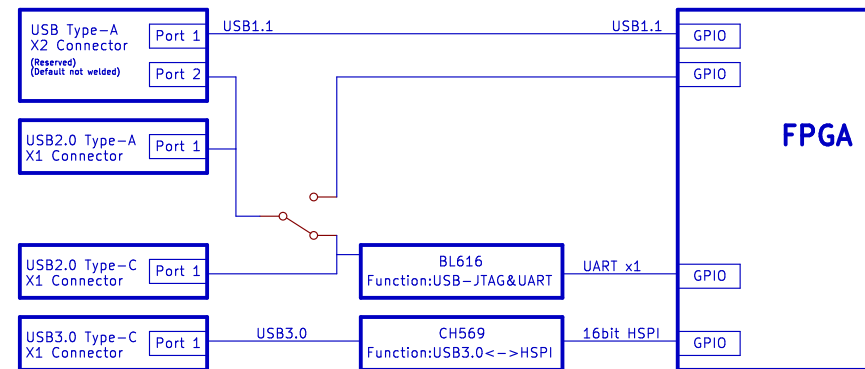
Rev: 1.4

Id: 4/21

IIC connection



USB connection



Sheet: /Block_Diagram/
File: Block_Diagram.kicad_sch

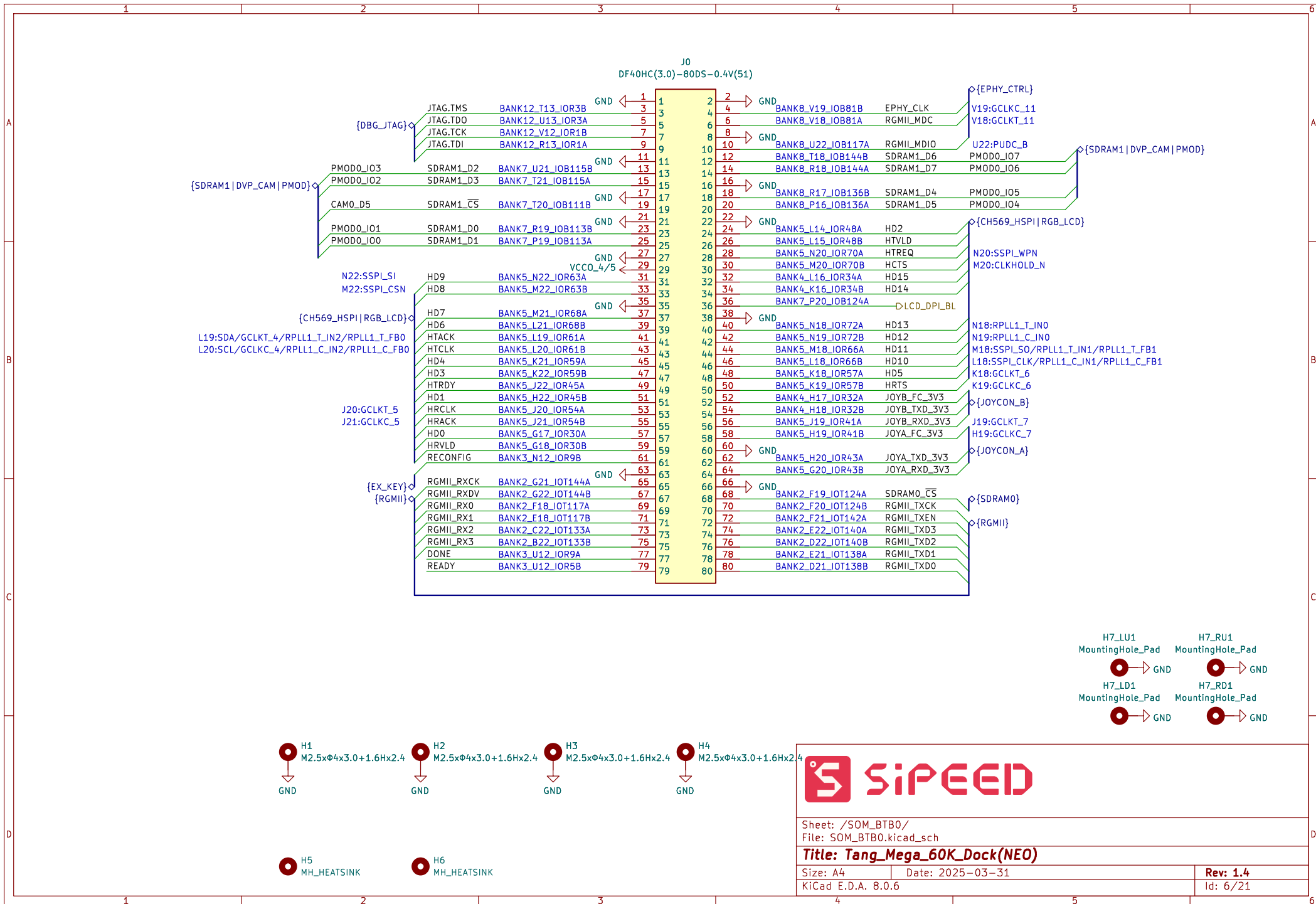
Title: Tang_Mega_60K_Dock(NEO)

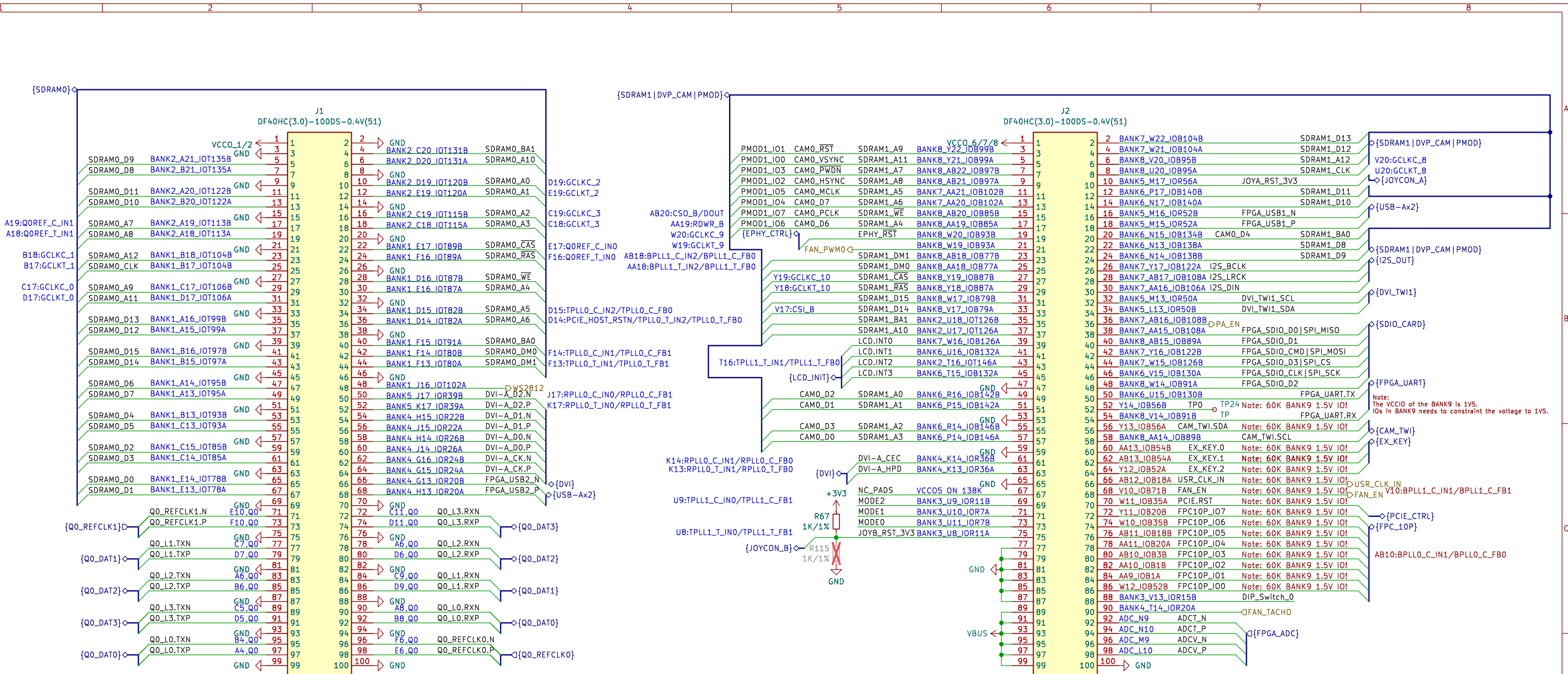
Size: A4 Date: 2025-03-31

KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 5/21





NOTE: Since the PINOUT of the SERDES part is not completely consistent with 138K, the Quad0 lanes 1 and 3 here are swapped relative to 138K, so PCIe can only work in x1 mode when SOM is 60K!!!

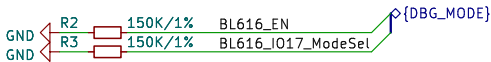
GW5AT-60 Configuration Modes

Configuration MODE	MODE[2..0]	Bus Width	Description
JTAG	XXX/101	—	GW5AT series of FPGA products are configured by external Host via JTAG interface
MSPI	001	x1, x2, x4	As a Master, FPGA reads data from external Flash (or other devices) via the SPI interface for configuration
SSPI	010	x1, x4	GW5AT series of FPGA products are configured by external Host via SPI interface
Master SERIAL	000	x1	Before FPGA used as a Slave, FPGA reads data from external devices via the DIN interface for configuration
Slave SERIAL	111	x1	GW5AT series of FPGA products are configured by external Host via DIN interface
Master CPU	100	x8, x16	Before FPGA used as a Slave, FPGA reads data from external devices via the DBUS interface for configuration
Slave CPU	110	x8, x16	GW5AT series of FPGA products are configured by external Host via DBUS interface

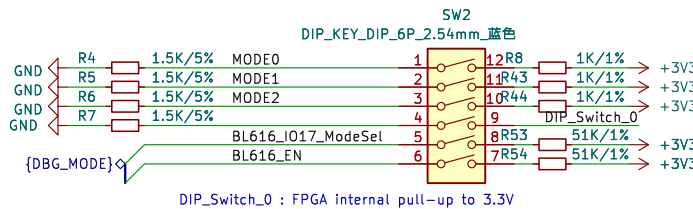
Please refer to GOWIN Manual UG718/UG718E for more details.

BL616 SEL[0..1] SWITCH

NO.	CONFIG	SEL0(EN)	SEL1(MODE)
0	Disable BL616 Use other debugger	0	X
1	JTAG mode	1	1
2	Game mode	1	0



Switch



Sheet: /SOM_BT1&2/
File: SOM_BT1-2.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

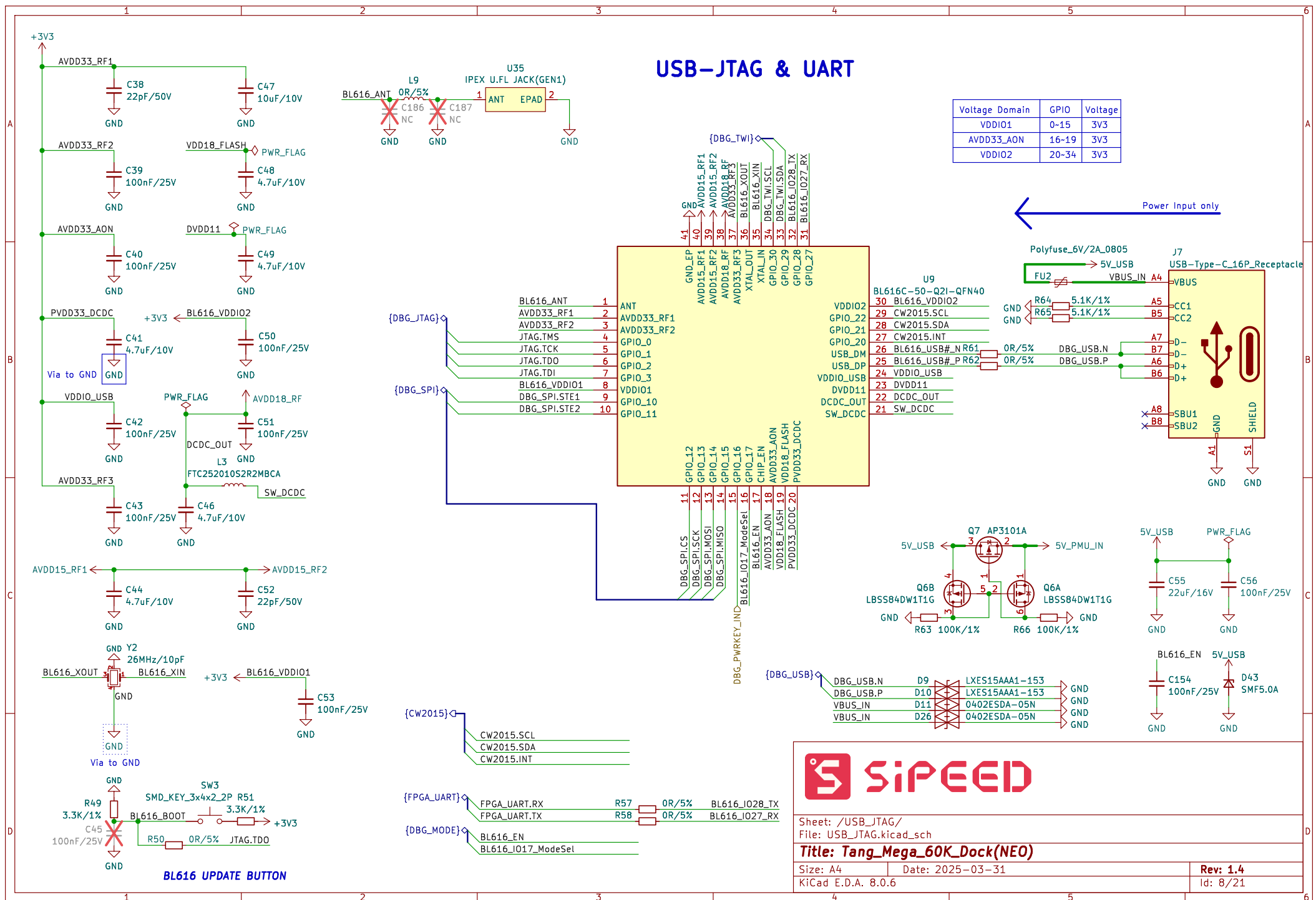
Size: A3

Date: 2025-03-31

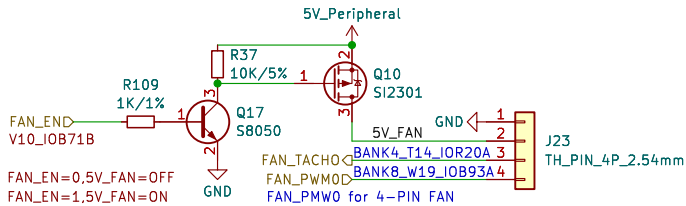
Rev: 1.4

KiCad E.D.A. 8.0.6

Id: 7/21



FAN Connector

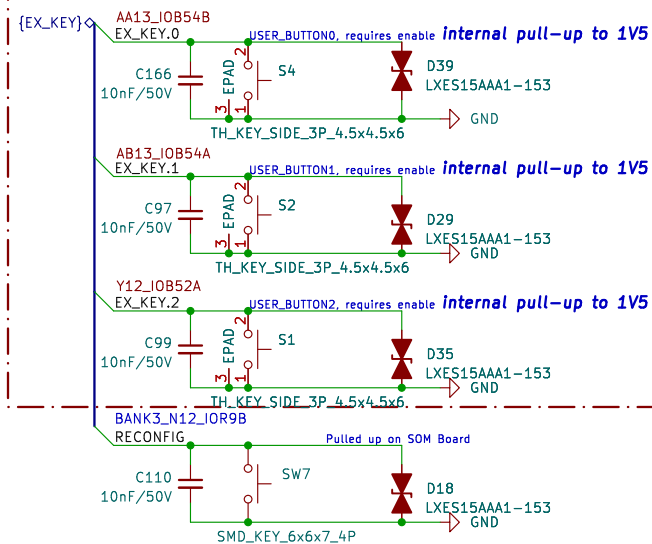


The IO of the **FAN.EN** is located in BANK9.
Note that the voltage needs to be constrained to 1V5.
IO_TYPE=LVC MOS15 PULL_MODE=NONE DRIVE=8 BANK_VCCIO=1.5

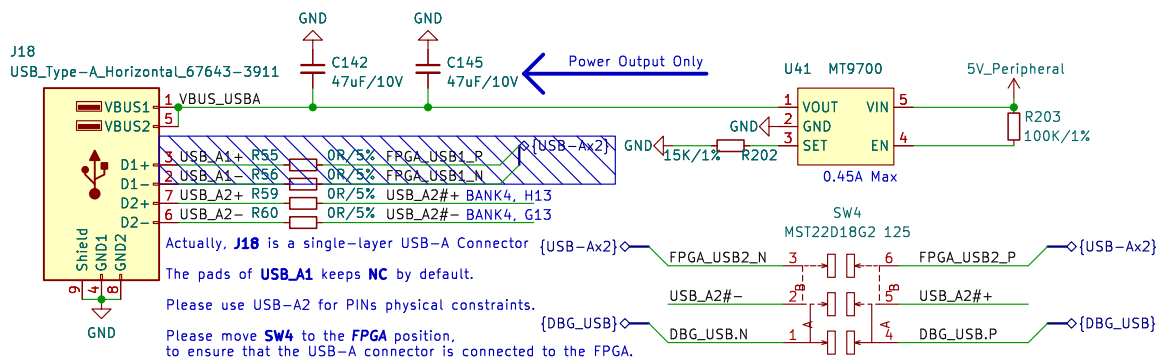


BUTTON

The IOs of these **BUTTONS** is located in BANK9. Note that the voltage needs to be constrained to 1V5.
IO_TYPE=LVC MOS15 PULL_MODE=UP DRIVE=OFF BANK_VCCIO=1.5



USB_A_x2

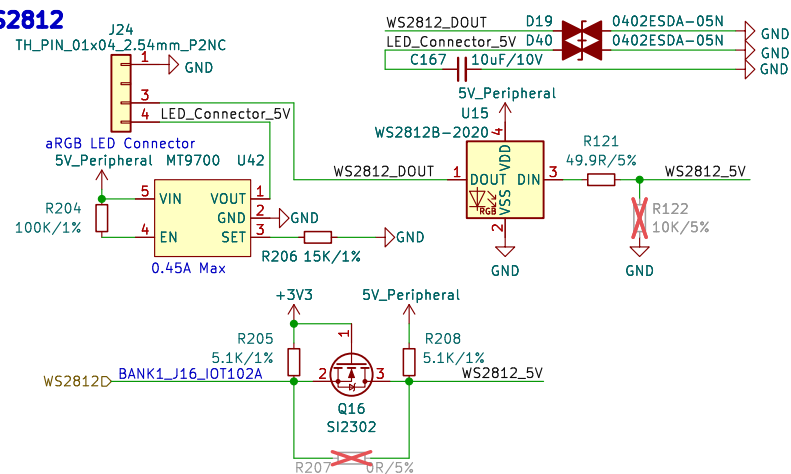


If SW4 is kept in **DBG** position, USB-A will be connected in parallel with the USB tracks of the debugger (BL616)!!! Please avoid leaving SW4 in the DBG position, otherwise this will cause USB communication issues with the debugger.

The only use case for putting SW4 in DBG position is to use BL616 as USB host, USB-A is more convenient for debugging than USB-C but since Bouffalo has given up maintaining its SDK, this feature will probably never be used.



WS2812



Sheet: /PERIPHERALS/
File: PERIPHERALS.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

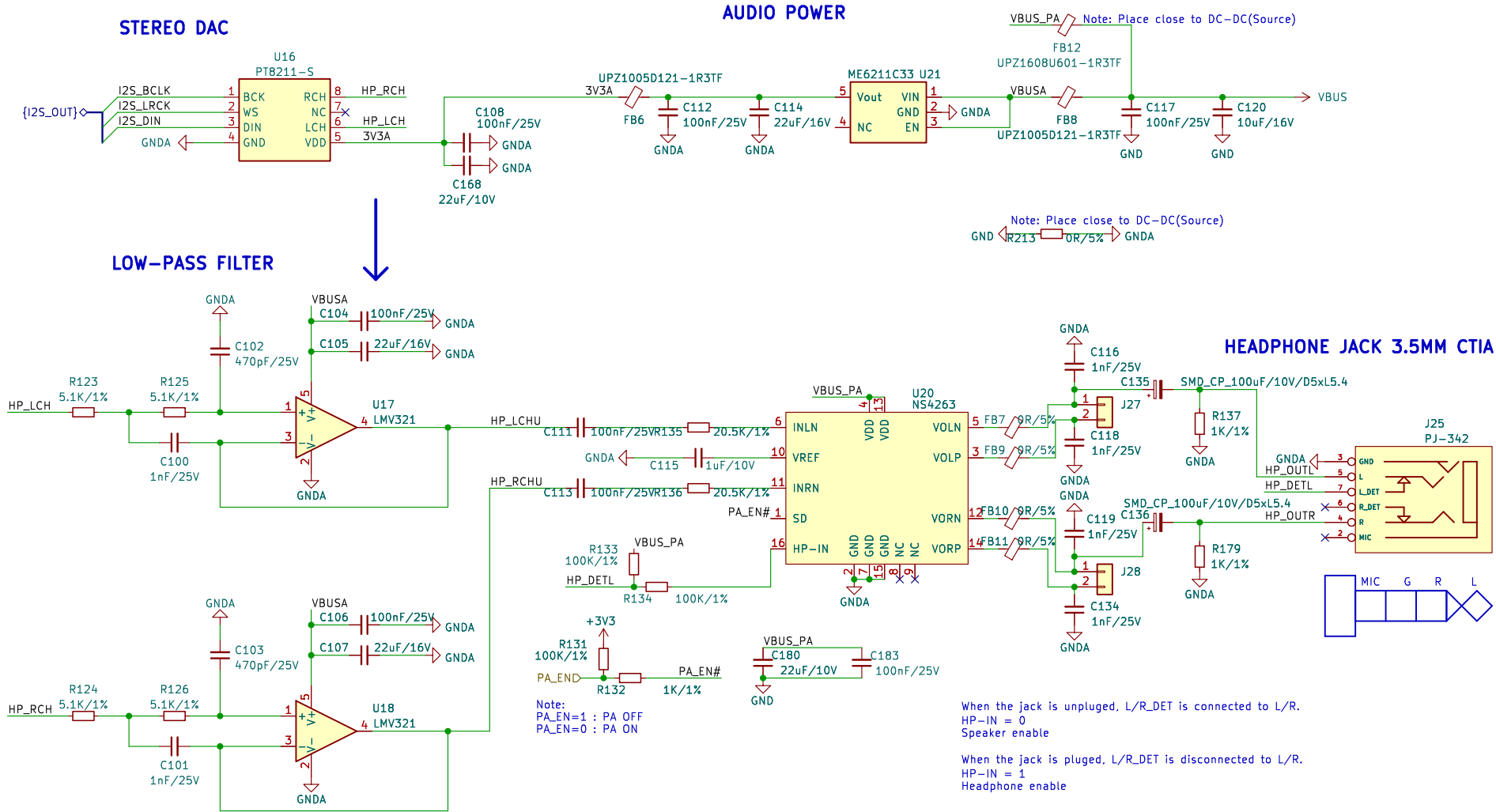
Size: A4	Date: 2025-03-31
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KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 9/21

FPGA–Audio



Sheet: /FPGA_AUDIO/
File: DAC+PA.kicad_sch

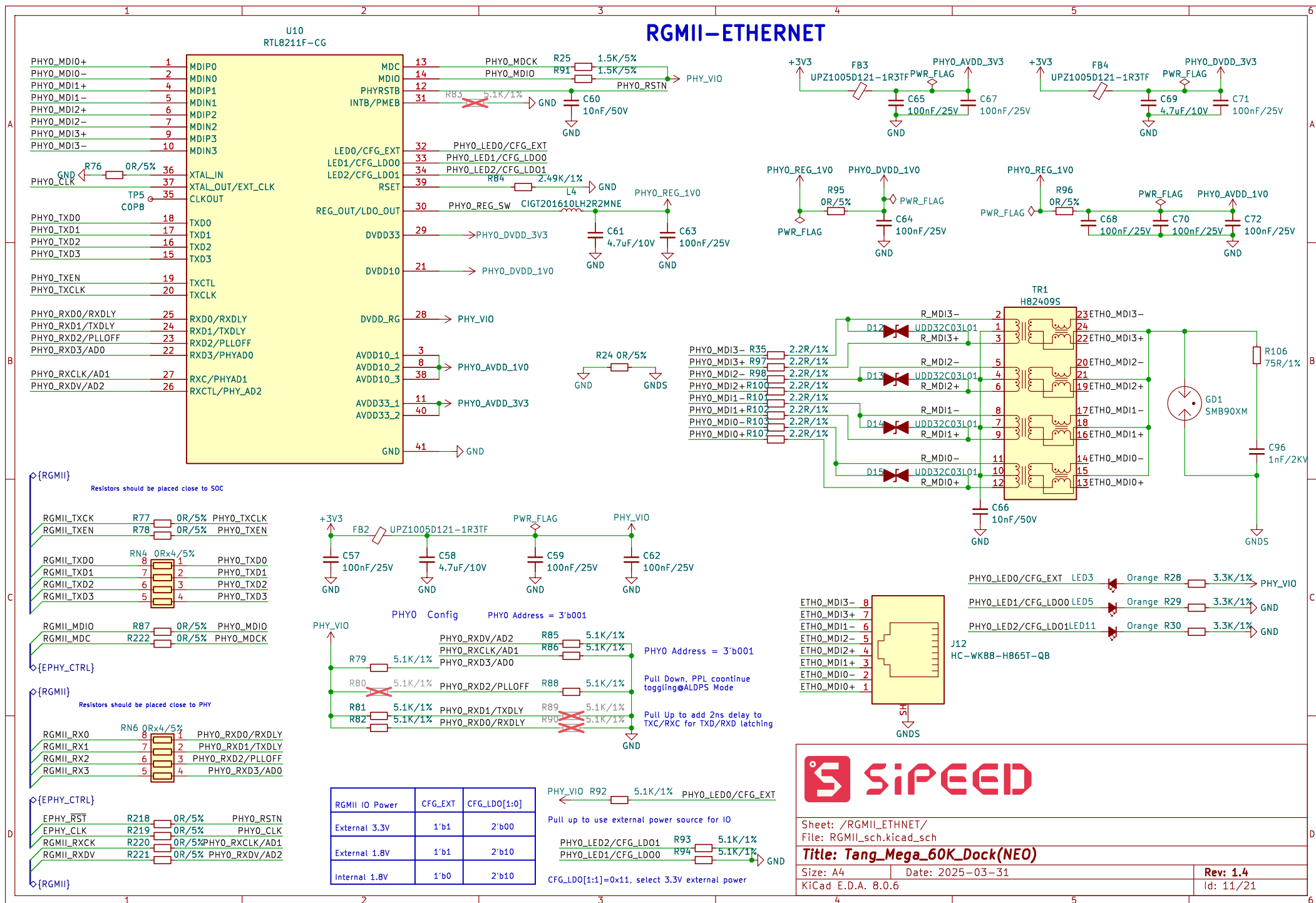
Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31
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KiCad E.D.A. 8.0.6

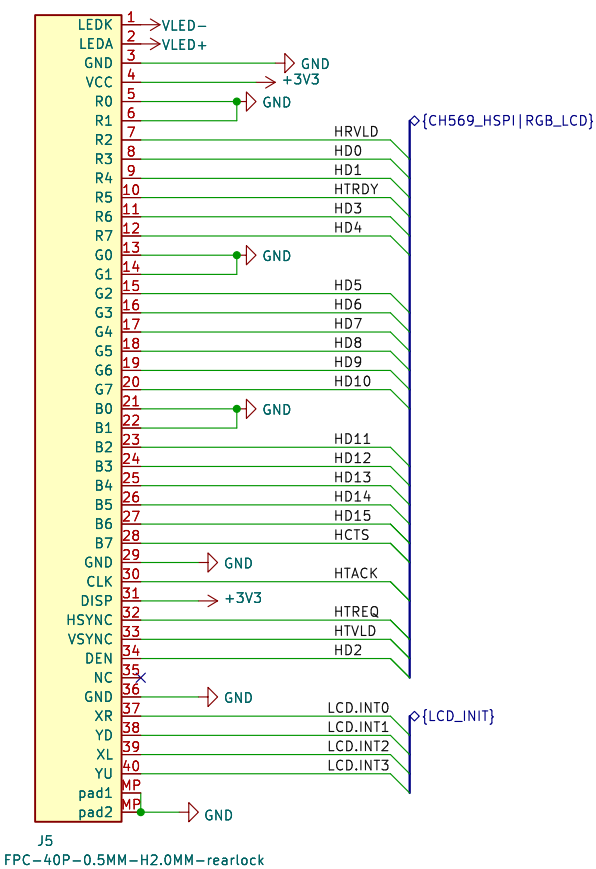
Rev: 1.4

Id: 10/21

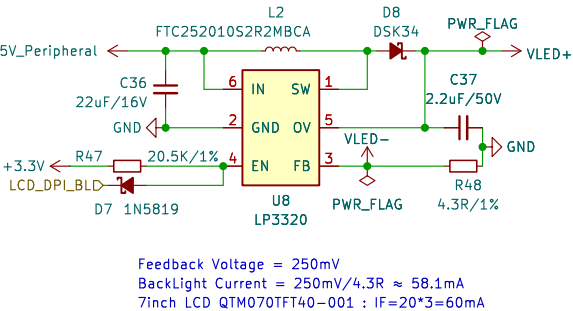


RGB LCD Display

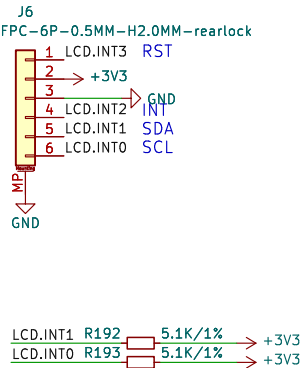
RGB LCD Display Connector



BACKLIGHT Driver



Capacitive Touch Connector



Sheet: /SYS_DPI/

File: LCD_DPI.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4

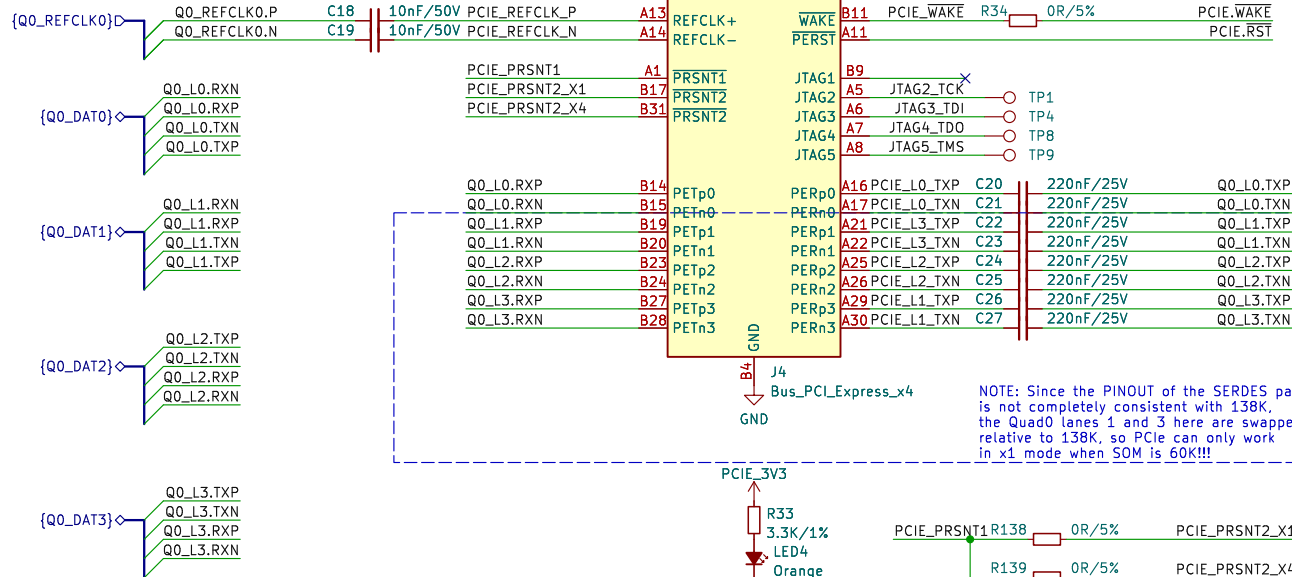
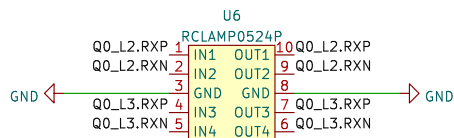
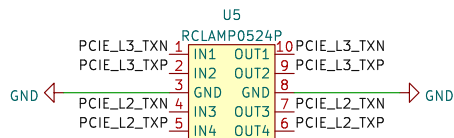
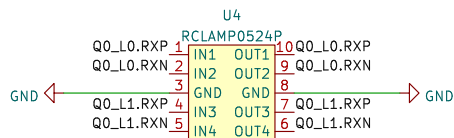
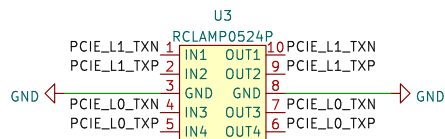
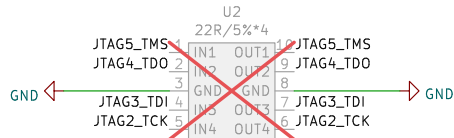
Date: 2025-03-31

KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 12/21

PCI-Express X4



Note:
PCIe control FPGA
FPGA internal pull-up to 1.5V(60K)
The IO of the **PCIE_RST** is located in BANK9.
Note that the voltage needs to be constrained to 1V5.
IO_TYPE=LVC MOS15 PULL_MODE=UP DRIVE=OFF BANK_VCCIO=1.5



Sheet: /PCIE_GT/
File: SERDES_PCIE.kicad_sch

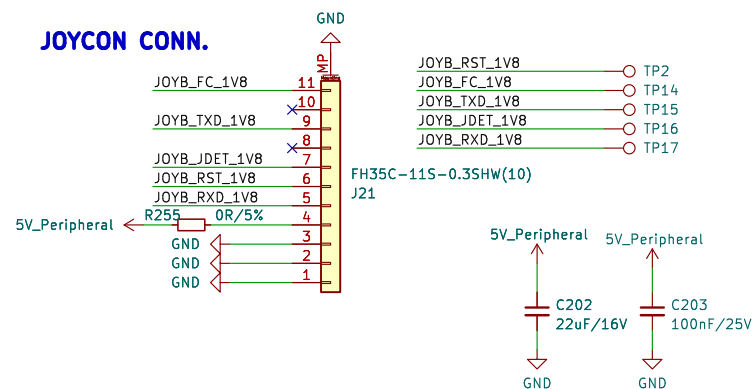
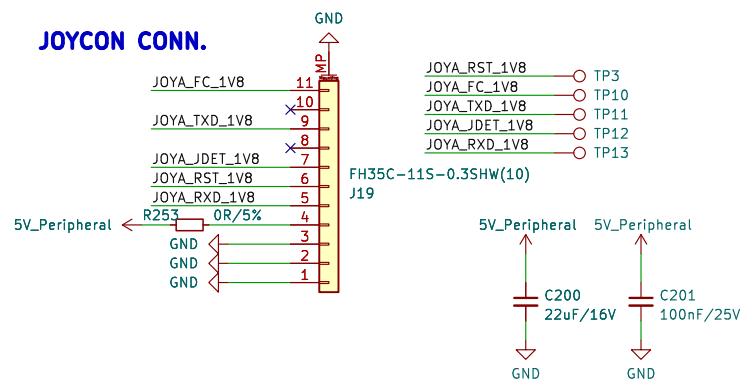
Title: Tang_Mega_60K_Dock(NEO)

Size: A4 Date: 2025-03-31

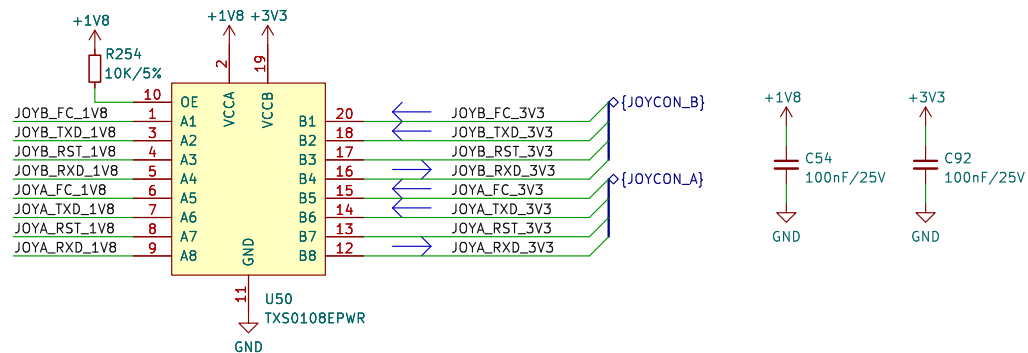
KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 13/21



Bi-directional Voltage Translator



Note:
RST=0: Enable JOYCON
RST=1: Disable JOYCON

JDET=1:
HIGH when connected to console via Bluetooth.
Joy-Con will not send serial data post-handshake unless pin is pulled LOW by console.



Sheet: /JOYCON/
File: JOYCON.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

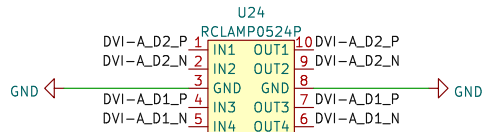
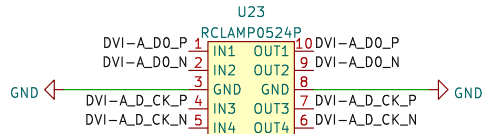
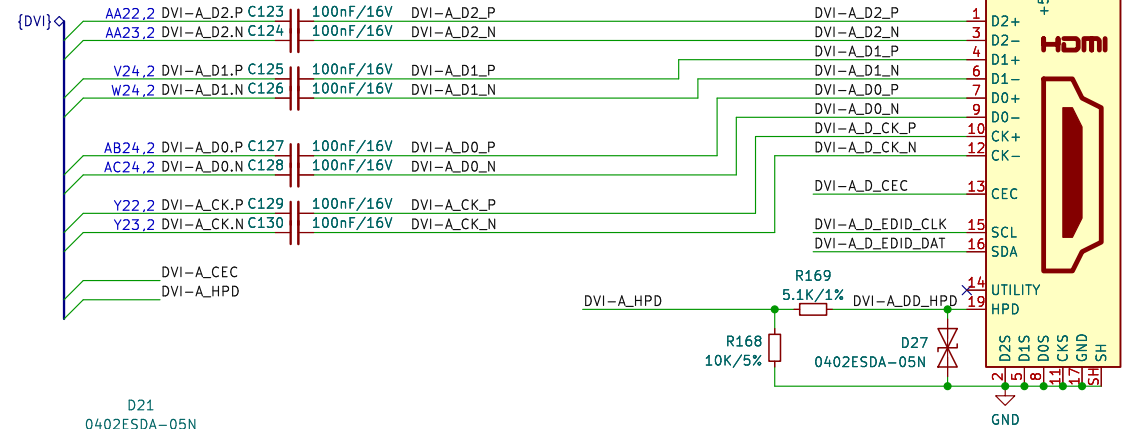
Size: A4 Date: 2025-03-31

KiCad E.D.A. 8.0.6

Rev: 1.4

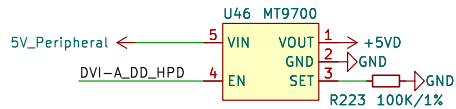
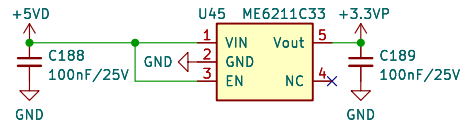
Id: 14/21

DVI(Only FOR TX ON 60K)



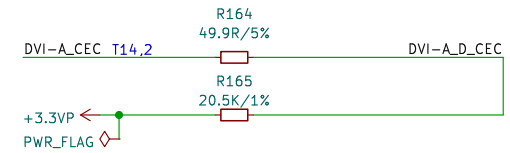
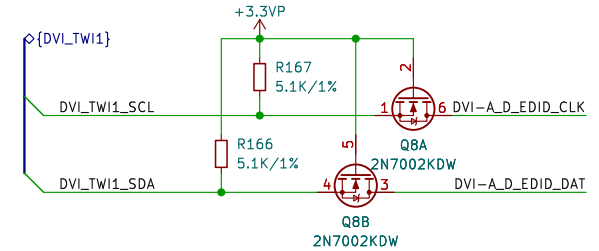
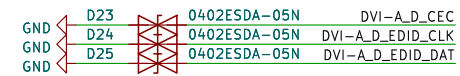
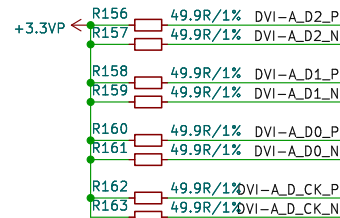
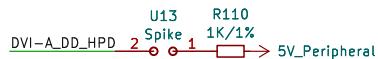
DVI Power control

(Save Battery Power)



Insert HDMI : HPD=1
Unplug HDMI : HPD=0

Current limit: 68mA



Sheet: /FPGA_TMD5_A/

File: DVI_A.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4

Date: 2025-03-31

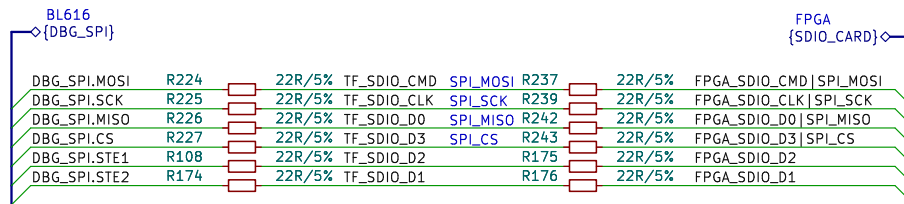
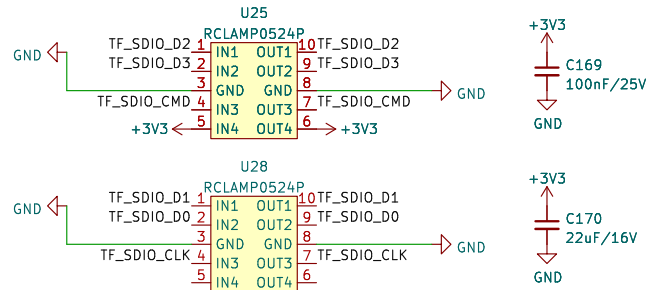
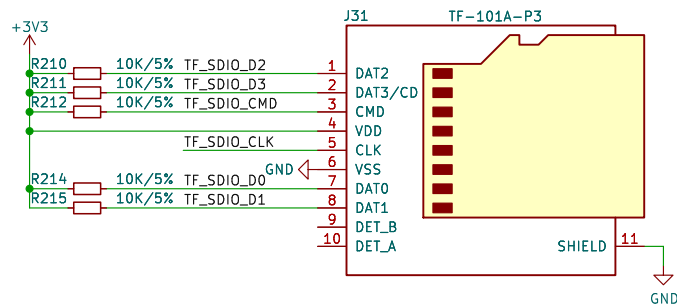
KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 15/21

STORAGE

BL616 MICRO SD CARD

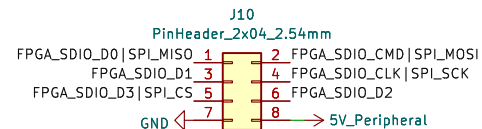


EVB mode : BL616_SPI high-Z , FPGA can work in SPI mode or SDIO mode

Game mode : BL616_SPI Master , FPGA_SPI Slave , FPGA_SPI_CS diasble , FPGA_SDIO_D1/D2 work as SPI_CS(input) or other function

Q1 : How to switch between EVB mode and Game mode?

A1: Switch DBG_ModeSel through SW2(6P Switch DIP)



Sheet: /SYS_STORAGE/

File: SDIO&M.2.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4

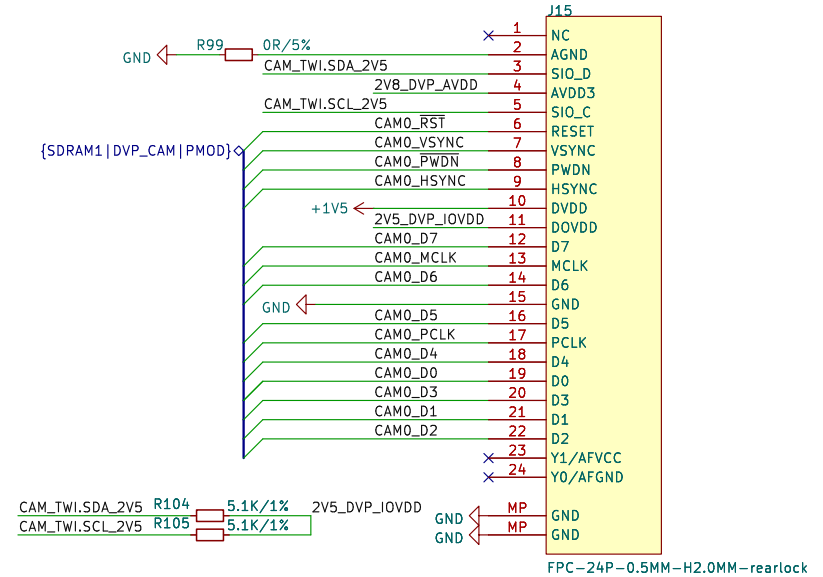
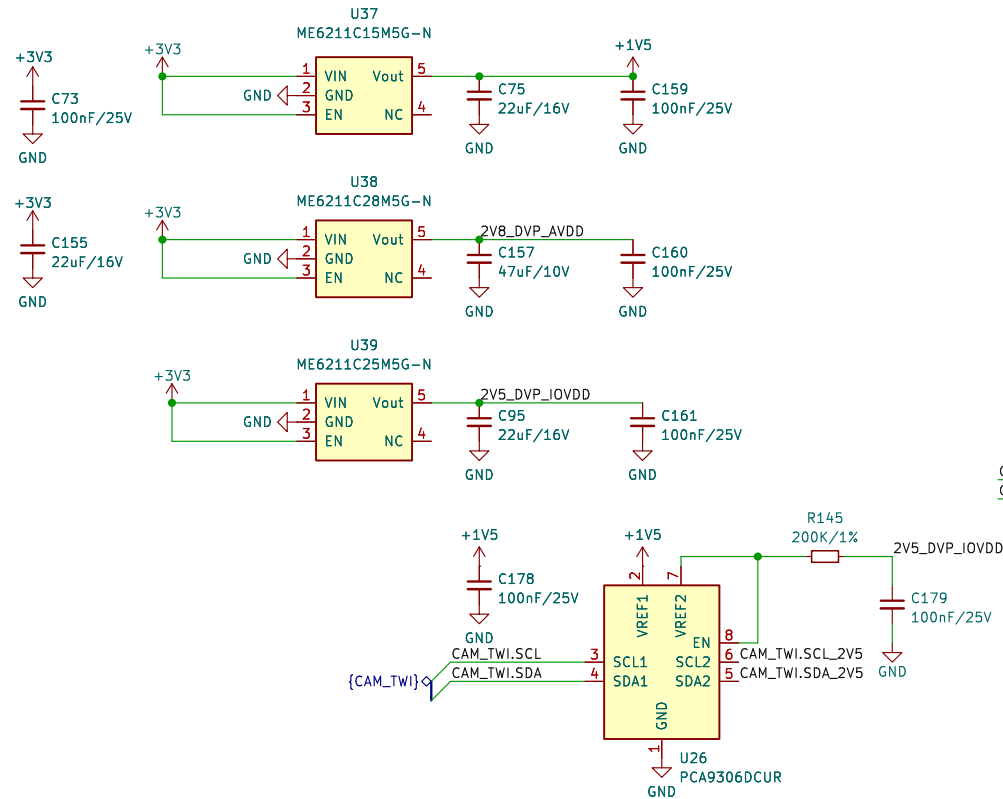
Date: 2025-03-31

Rev: 1.4

KiCad E.D.A. 8.0.6

Id: 16/21

DVP-CAMERA



Sheet: /DVP_CAMERA/
File: DVP_CAM.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4 Date: 2025-03-31

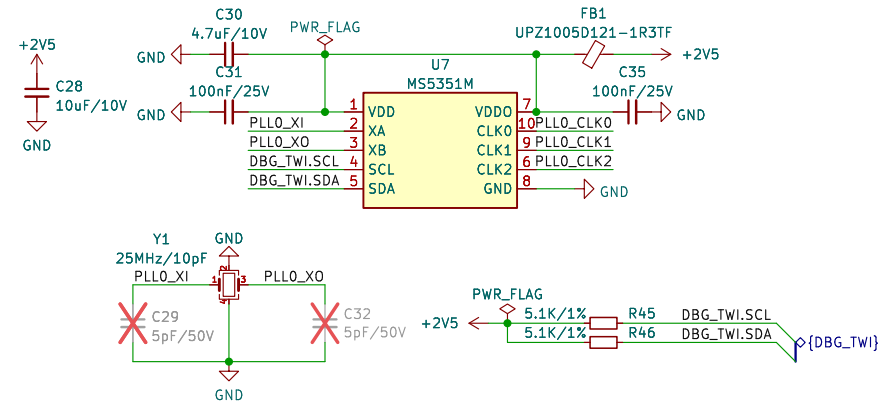
KiCad E.D.A. 8.0.6

Rev: 1.4

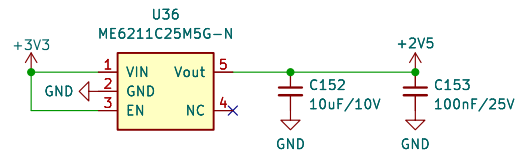
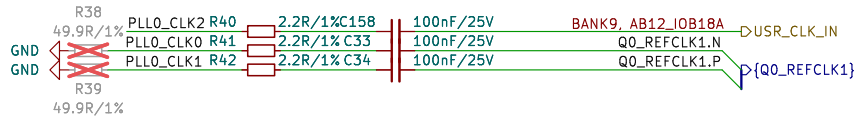
Id: 17/21

CLOCK

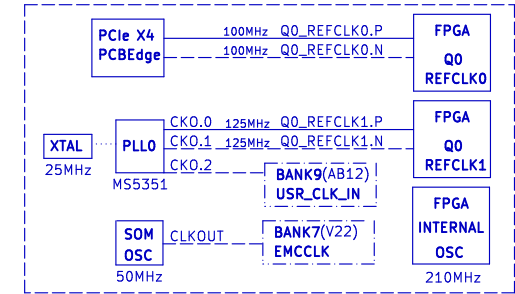
EXT. PLL



The IO of **USR_CLK_IN** is located in BANK9. Note that the voltage needs to be constrained to 1V5.
IO_TYPE=LVC MOS15 PULL_MODE=DOWN DRIVE=OFF BANK_VCCIO=1.5



COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of the option "Frequency Divider" to generate a different output clock frequency, which can be one of 64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10, or 1/126 of the oscillator frequency.



Sheet: /SYS_CLOCK/
File: CLOCK.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4 Date: 2025-03-31

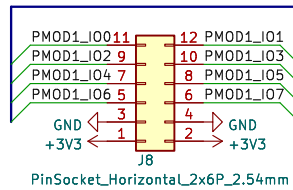
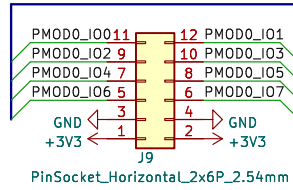
KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 18/21

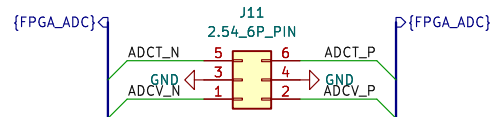
EX CONN.

PMOD

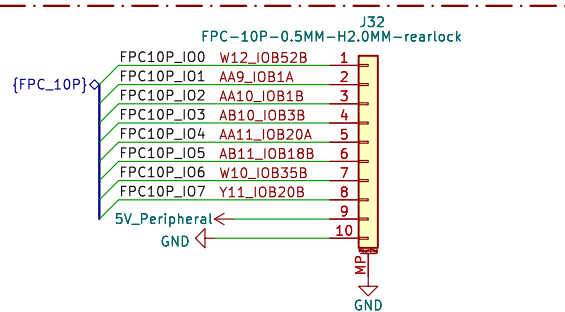


{SDRAM1 | DVP_CAM | PMOD}

ADC CONN.

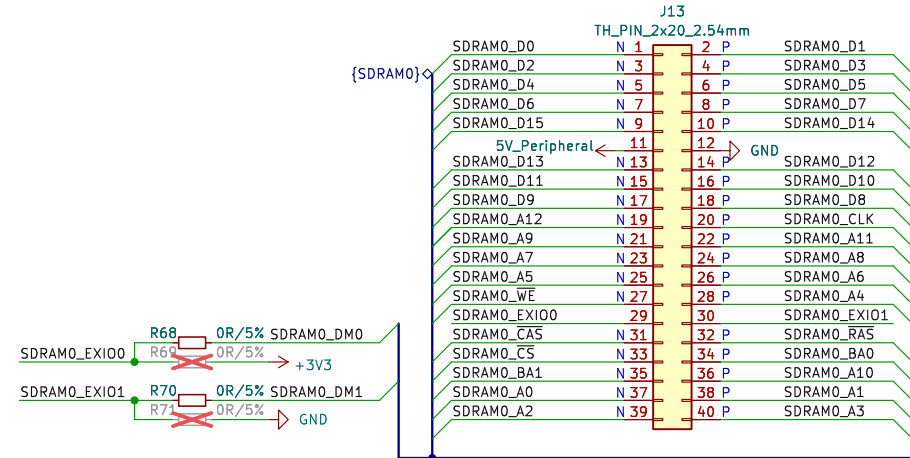


FPC10P CONN.

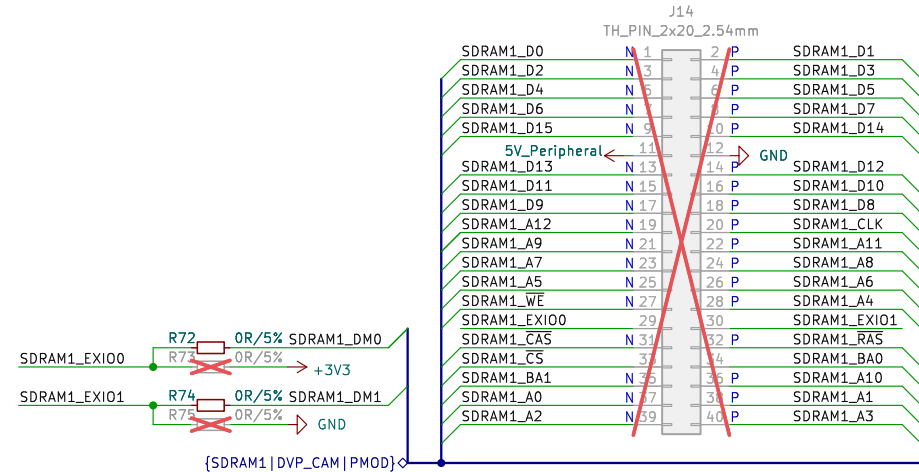


The IOs on the FPC-10P is located in BANK9. Note that the voltage needs to be constrained to 1V5.

SDRAM0 CONN. Close to PCIe connector



SDRAM1 CONN. Close to PMOD connector



Sheet: /FPGA_EXT_CONN./
File: EX_CONN.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

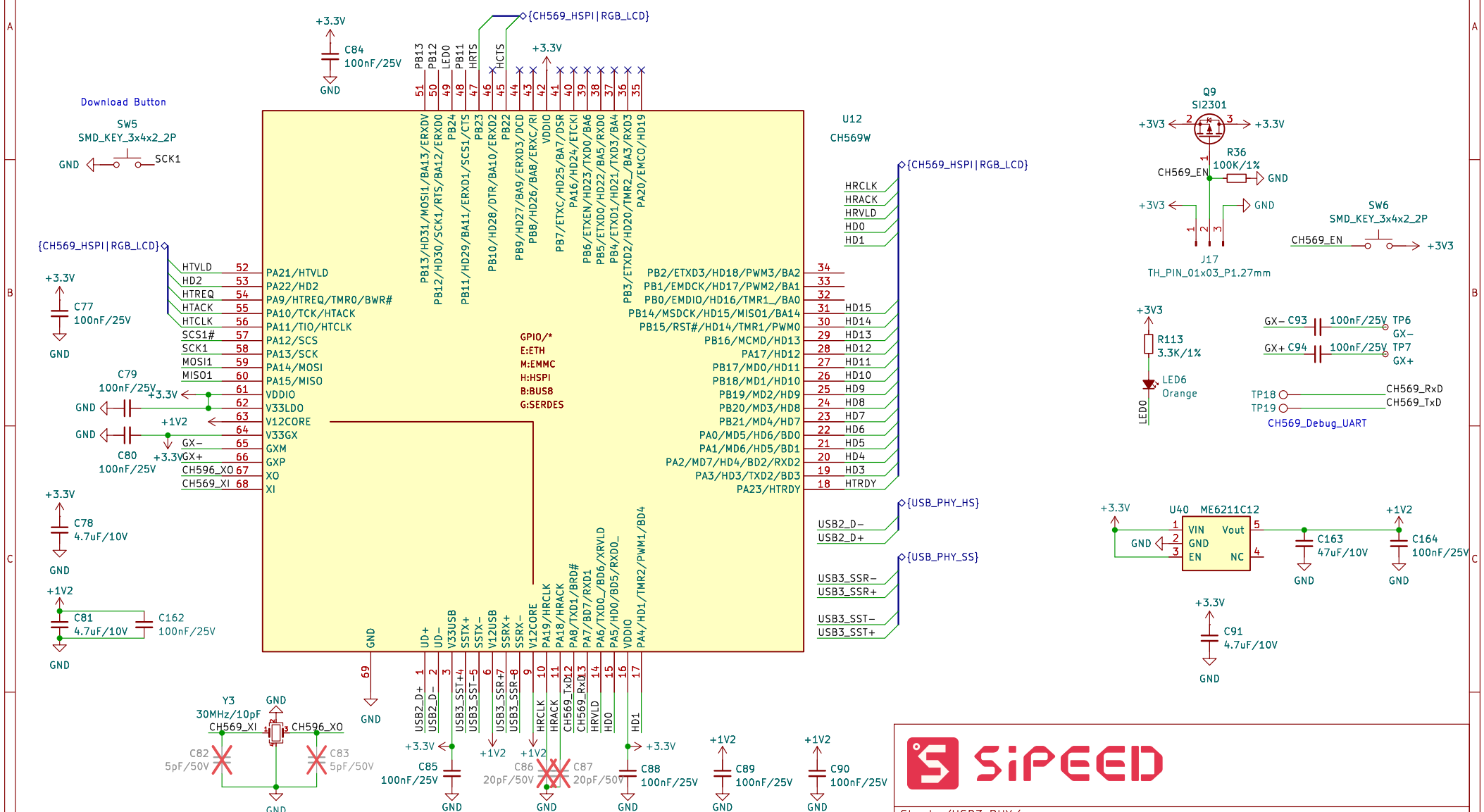
Size: A4 Date: 2025-03-31

KiCad E.D.A. 8.0.6

Rev: 1.4

Id: 19/21

USB3-PHY



Sheet: /USB3_PHY/

File: USB3_PHY.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4

Date: 2025-03-31

Rev: 1.4

Id: 21/21